

GenAI Amplifier

for Software &
Quality Engineering

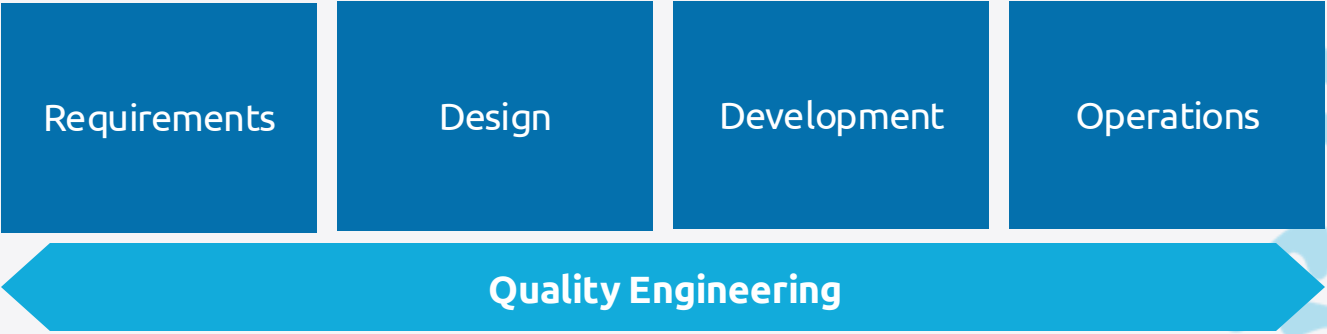


Section 1

Evolution

What History tells us

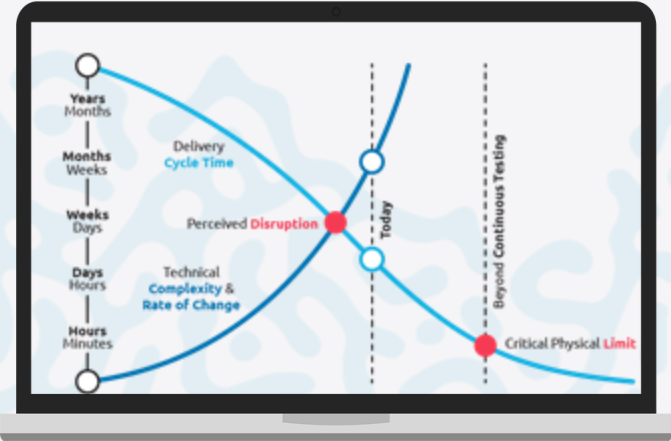
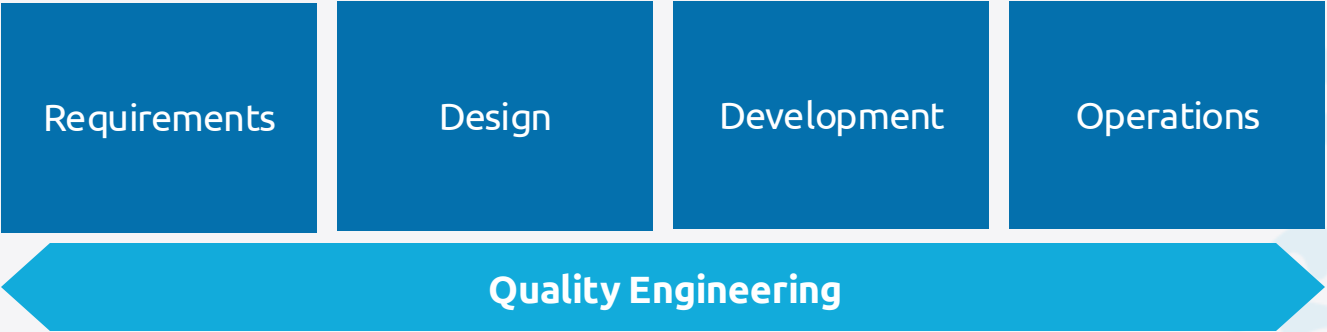
Inspection era	Formalization era	Agile and Automation era
- Reactive testing focused on debugging. - Responsibility was isolated within development teams - Basic, manual techniques like print statements, with no formal testing tools	- Testing became structured within QA teams, integrated into the SDLC. - Standards like IEEE 829 and CMM formalized processes - The rise of test automation	- Quality Engineering embedded in Agile and DevOps - Mobile shifted focus to user experience - Continuous Testing with automation, low-code, and the rise of AI tools
1940s-70s	1980s-90s	2000s-10s
<i>"Testing shows the presence, not the absence of bugs." ¹</i>	<i>"The more you test software, the more immune it becomes to your tests." ²</i>	<i>"If you automate a mess, you get an automated mess." ³</i>



1 - Edsger Dijkstra, 1969
2 - Boris Beizer, 1990
3 - Rod Michael, 2017

What History tells us for our Present

Inspection era	Formalization era	Agile and Automation era	AI-driven quality era
- Reactive testing focused on debugging. - Responsibility was isolated within development teams - Basic, manual techniques like print statements, with no formal testing tools	- Testing became structured within QA teams, integrated into the SDLC. - Standards like IEEE 829 and CMM formalized processes - The rise of test automation	- Quality Engineering embedded in Agile and DevOps - Mobile shifted focus to user experience - Continuous Testing with automation, low-code, and the rise of AI tools	- Generative AI amplifies human-driven quality engineering - AI-powered platforms complement and extend beyond human-designed testing <i>Focus on real-time feedback for risk, quality, and non-quality issues</i>
1940s-70s	1980s-90s	2000s-10s	2020s
<i>"Testing shows the presence, not the absence of bugs." ¹</i>	<i>"The more you test software, the more immune it becomes to your tests." ²</i>	<i>"If you automate a mess, you get an automated mess." ³</i>	<i>"It is amazing what you can do when you know what to ask for."</i>



1 - Edsger Dijkstra, 1969
2 - Boris Beizer, 1990
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What the Present is shaping for Tomorrow



Where is Generative AI making the most significant impact on Software Quality Assurance?

Higher productivity and speed in the quality engineering process makes it possible to ship code more frequently with a high customer experience.

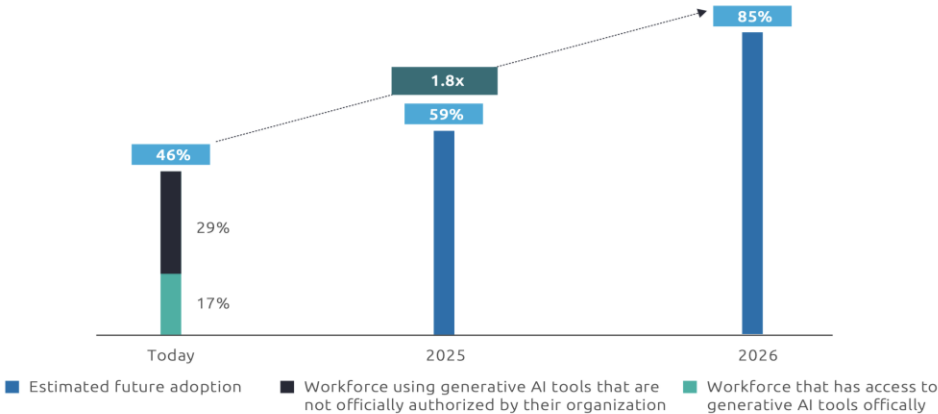
- 79%** Quality leaders agree that AI systems are going to be used to help them optimize their test scope and increase velocity¹.
- 78%** Quality leaders agree that they will use AI to speed up their test design¹.
- 73%** Quality leaders intend to use AI for test results analysis and defect prediction¹.

¹ Data from World Quality Report 15th Edition, 2023-24.
² Representing the value of generative AI: Top use cases across industries – Capgemini Research Institute.
Sogeti Gen AI Amplifier for Software & Quality Engineering

<https://www.capgemini.com/insights/research-library/world-quality-report-2023-24/>

Adoption of generative AI in software engineering is in its early stages, but more than four in five software professionals are estimated to leverage* it by 2026

Percentage of workforce leveraging generative AI tools in the workforce

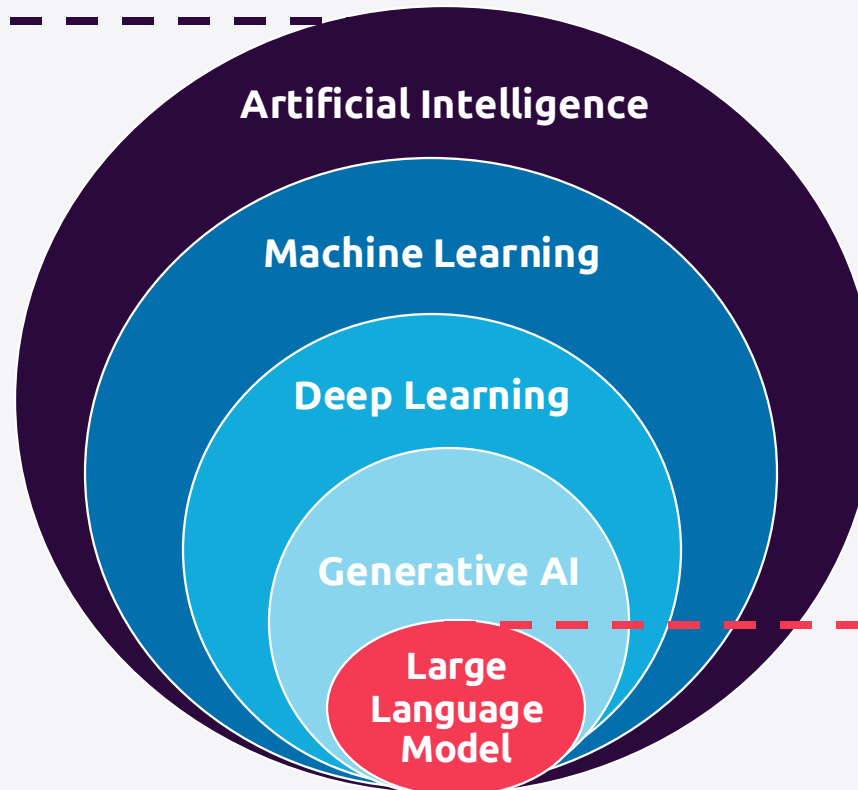


<https://www.capgemini.com/insights/research-library/gen-ai-in-software/>

Technical and analytical tasks — — — — —

Complex object detection
Visual recognition
self-healing automation tests
Root cause analysis
user flow validation
Predictive metrics
Synthetic data generation
...

<https://www.sogeti.com/ai-for-qe/>



— — — — — Natural language-driven tasks

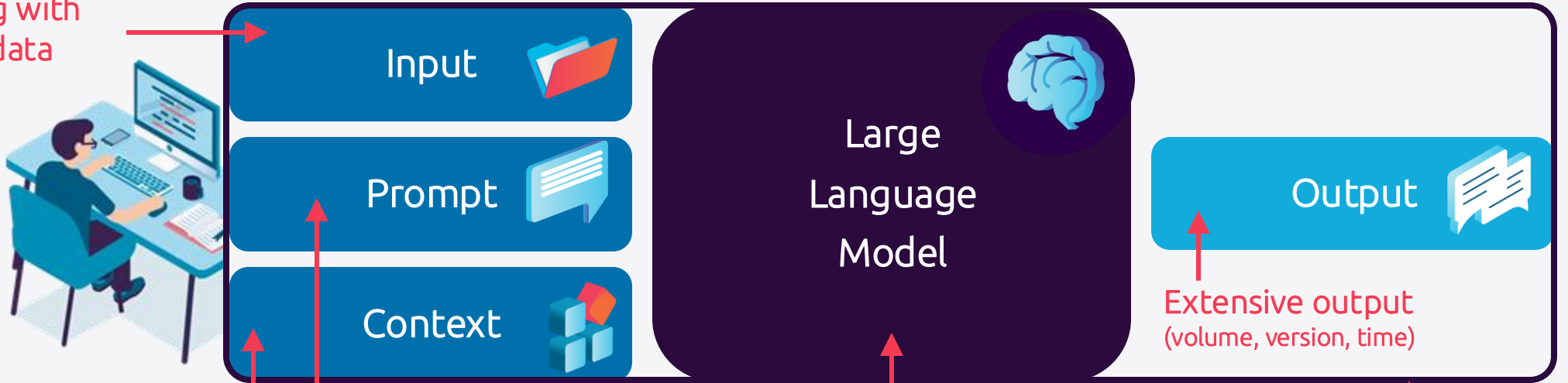


Section 2

How to generate value?

From using **Large Language Models**

Overloading with complex data



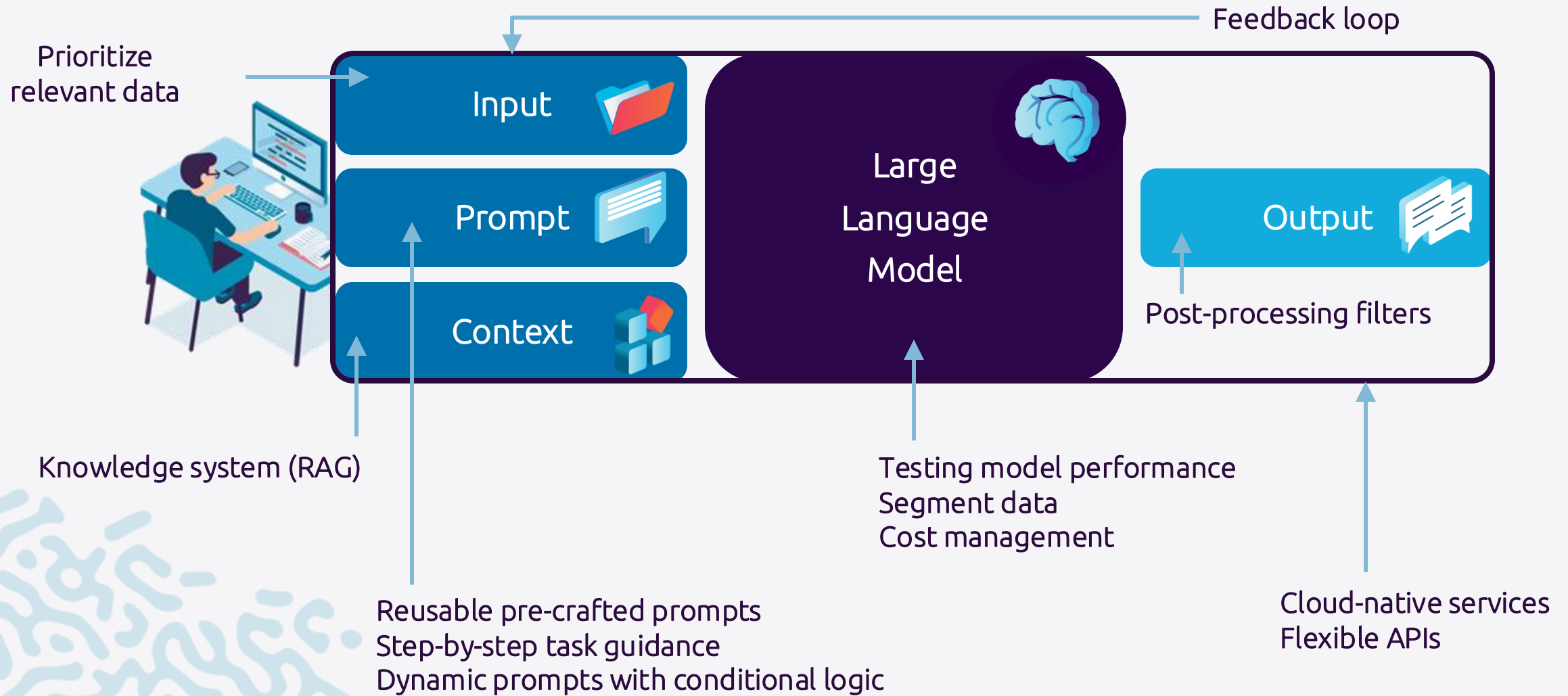
Misinterpretation

Lack of precision in prompts
Lack of reusability of prompts
Insufficient explicit instructions
Endless prompt for unique task

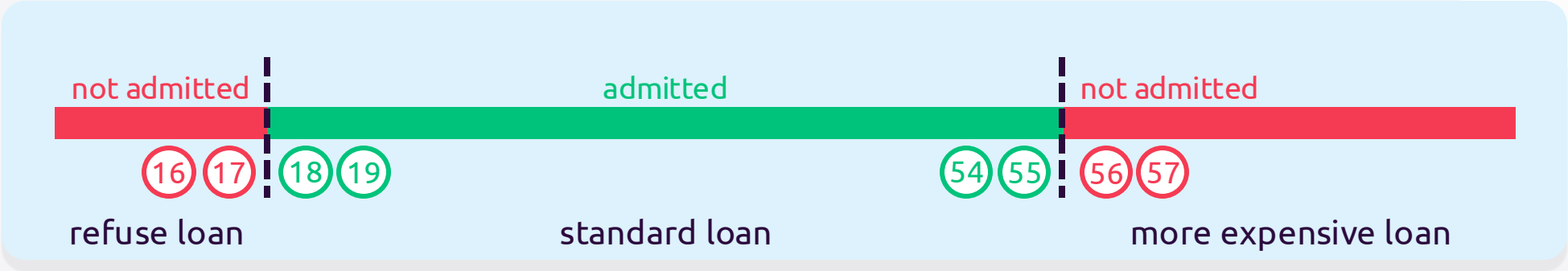
Model performance
Context windows limit
Cost

Lack of data privacy
Limited integration

From using Large Language Models to generating **Value**



Input:



Prompt: “Generate test cases applying 4-value Boundary Value Analysis. As table. No need for explanation.”

Output:

LLM 1

Here is a table applying the 4-value boundary value analysis for the scenario depicted in the image:

Test Case #	Input Value	Expected Outcome
1	17	Not Admitted (Refuse Loan)
2	18	Standard Loan Admitted
3	55	Standard Loan Admitted
4	56	Not Admitted (More Expensive Loan)

These test cases target the boundaries between the refusal, standard, and more expensive loan categories.

2-value boundary

LLM 2

Test Case ID	Credit Score	Expected Result
TC01	16	Loan Refused
TC02	17	Loan Refused
TC03	18	Standard Loan Offer
TC04	55	Standard Loan Offer
TC05	56	More Expensive Loan Offer
TC06	57	More Expensive Loan Offer

3-value boundary

LLM 3

Test Case	Input Value	Expected Output
1	16	not admitted
2	17	not admitted
3	18	not admitted
4	54	standard loan
5	55	standard loan
6	56	standard loan
7	57	more expensive loan
8	100	more expensive loan

Incorrect 4-value boundary

LLM 4

Here are the test cases applying 4-value Boundary Value Analysis, presented as a table:

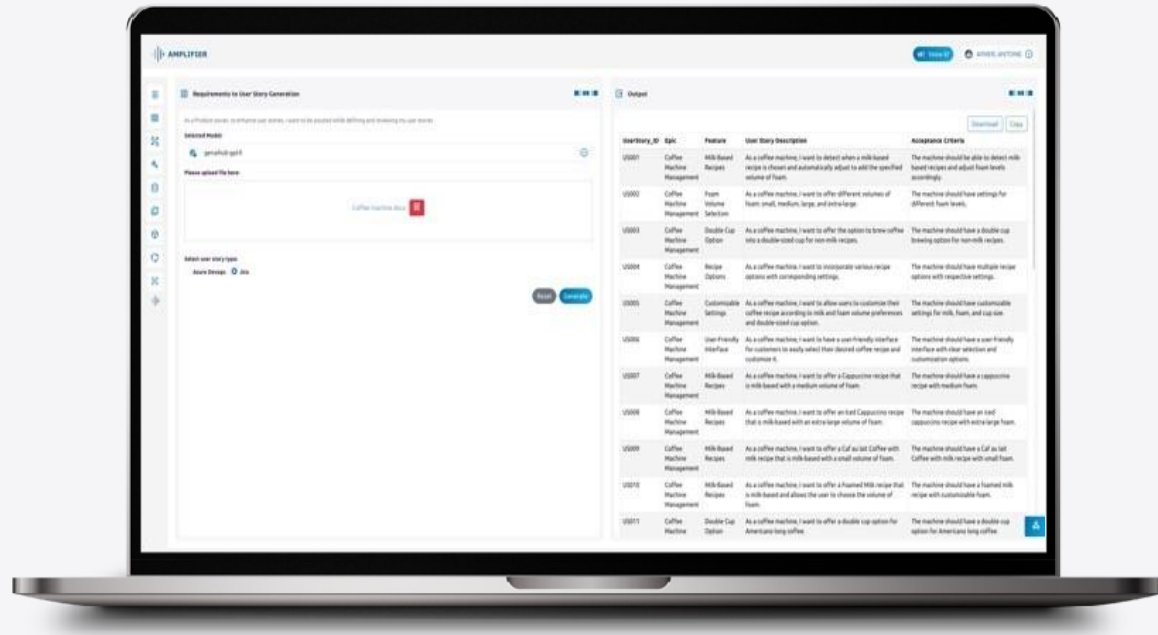
Test Case	Input Value	Expected Output
1	16	Refuse loan
2	17	Refuse loan
3	18	Standard loan
4	19	Standard loan
5	54	Standard loan
5	55	Standard loan
7	56	More expensive loan
8	57	More expensive loan

4-value boundary

Note: The outputs shown represent the most frequent results from identical instructions given to each public LLM, tested 10 times. As LLMs are frequently updated, these results may not reflect current and future performance.

Sogeti GenAI Amplifier accelerating software & quality engineering

Our *GenAI Amplifier for Software & Quality Engineering* enables faster delivery of critical steps across the software lifecycle from planning, design, build, test to deploy by using the power of multiple Gen AI systems.



>30 Prioritized Use Cases

from the inception of requirements and infrastructure design to software testing

Pre-crafted Prompts and Conversational

fine-tuned for nuanced responses

Safely Installed in Secure Cloud Environment

amplifying user workflows and system interactions

Built on a solid foundation with
our proprietary assets

Trusted and validated
for consistent, reliable results

Prebuilt prompts to minimize
variability and improve efficiency

With our secure Gen AI platform as foundation, we've integrated multiple Gen AI models, with pre-crafted prompts that are empowered with our industry-leading quality engineering methodology, agile framework and cloud-native architectural best-practices to ensure reliable high-quality outcomes.

Delivering uses cases across the lifecycle

Plan

- Requirement Creation
- Ambiguity Analysis
- Requirement Refinements
- Requirement Testability
- Requirement to User Story Generation

Build

- Design to Code Structure

Test

- Test Strategy
- Coverage-Based Testing (TMAP)
- Generic Testing
- Test Automation Design
- Test Data Management
- Infrastructure Testing
- SAP testing

Refactor

- Code generation
- Code Transformation
- UI Component

1



2



3



4



5



6



7



8



Design

- Requirement to Design
- Requirement to Data Model

Integration

- API Integration
- Integration Script generation

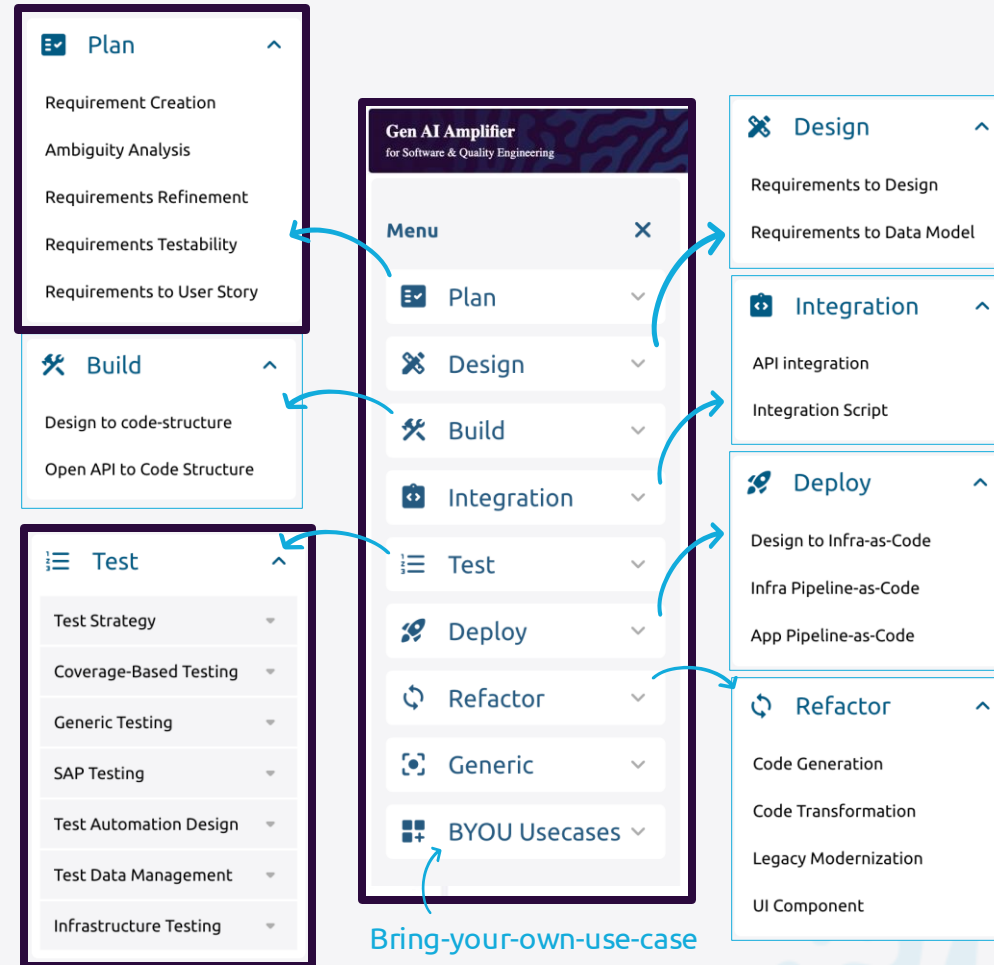
Deploy

- Design to Infra-as-Code
- Application Pipeline-as-Code
- Infra Pipeline-as-Code

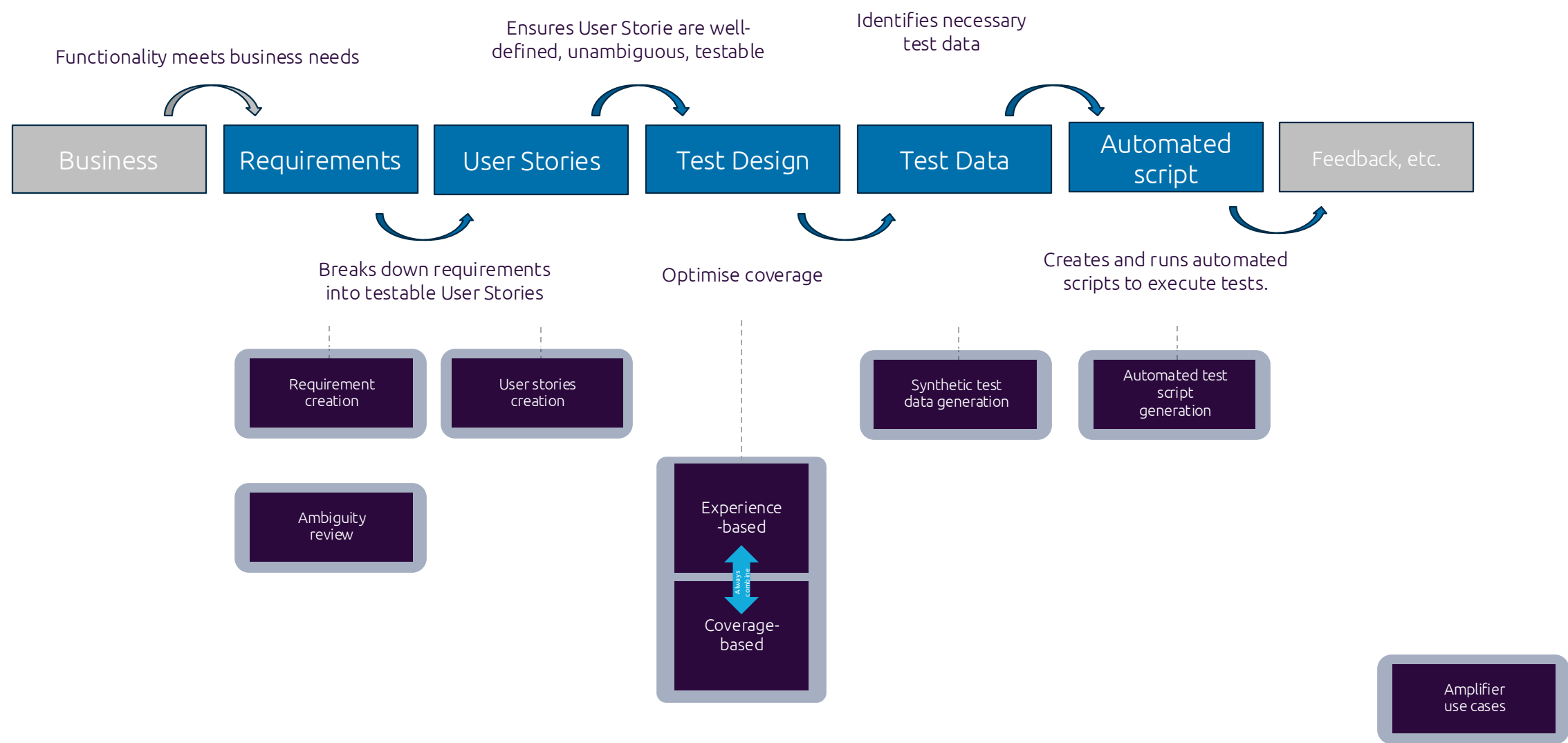
Bring-your-own-use-case

Gen AI Amplifier for Software & Quality Engineering

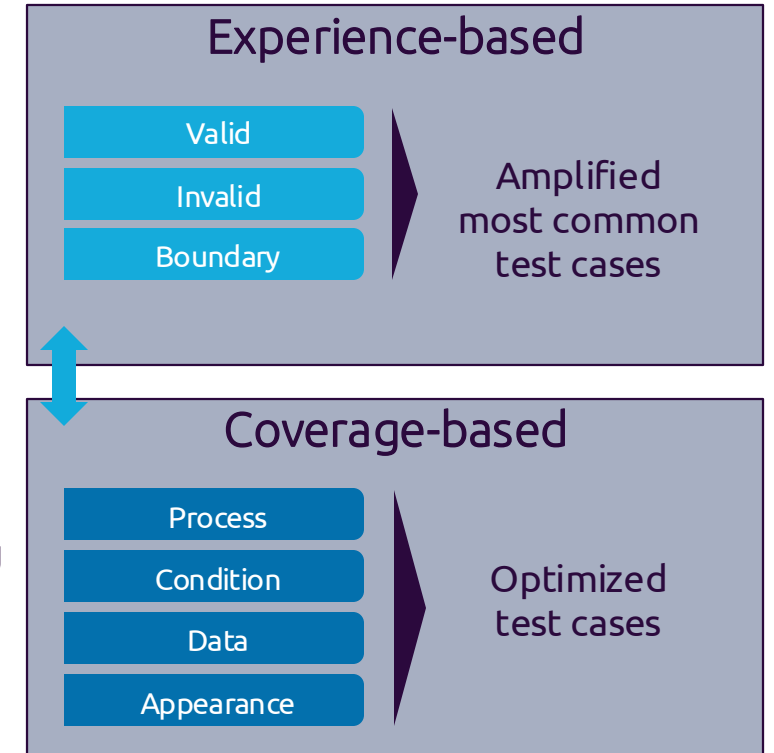
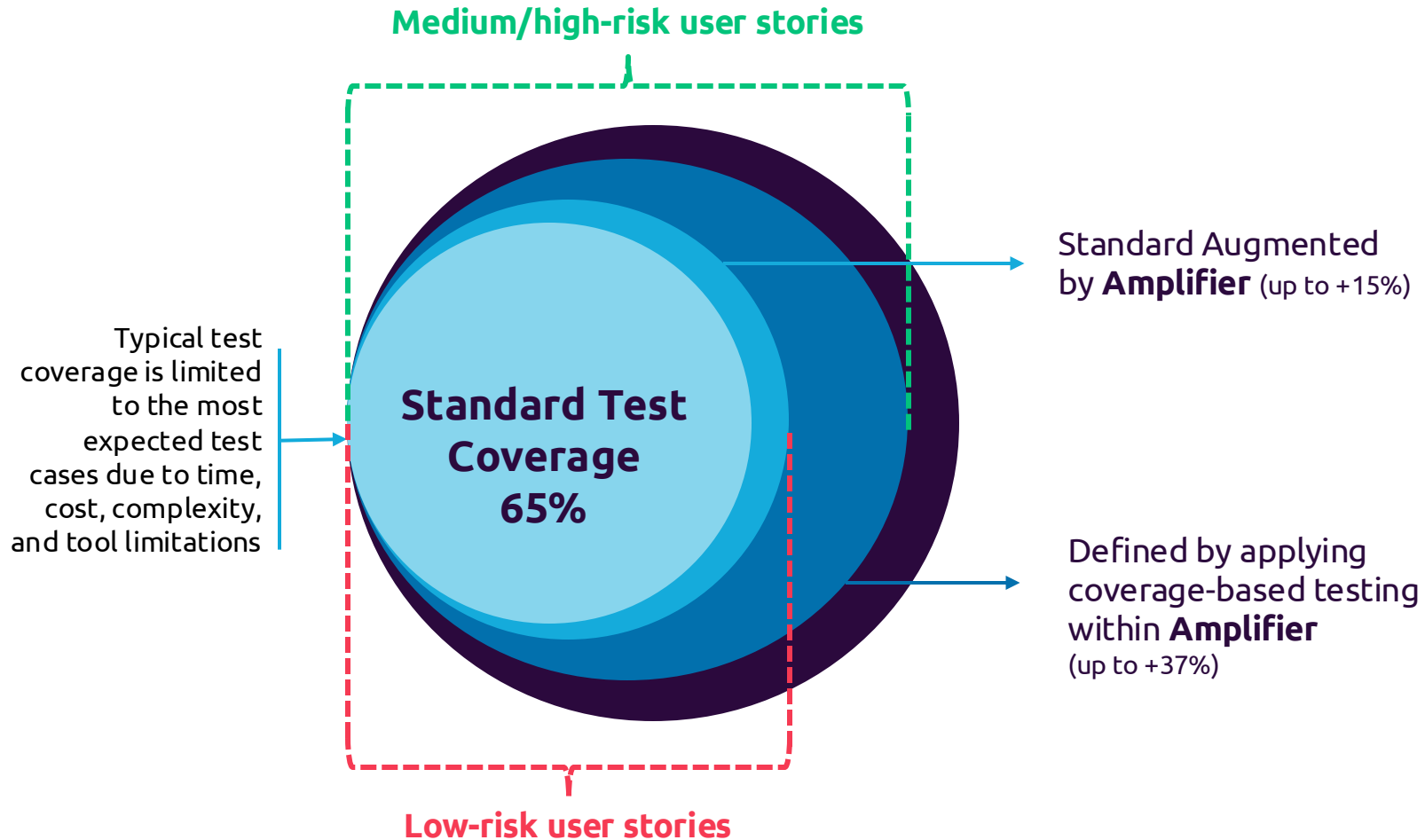
Homegrown Accelerator of Services



A journey inside the Amplified process



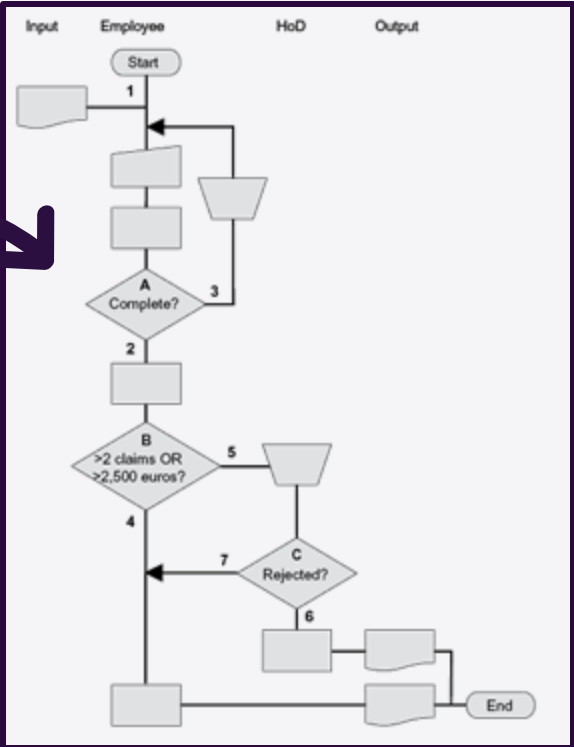
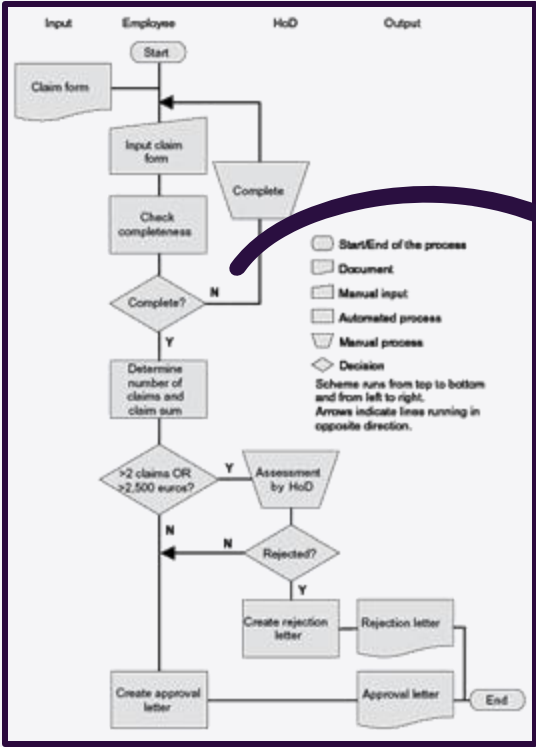
The amplified functional tester



<https://www.tmap.net/building-blocks/test-design-techniques>

Process-driven testing: Process Cycle Test

Verify that individual actions within a procedural flow can be executed correctly by covering **all variations** in processing paths and decision points.



The tester has described the following actions in respect of TC3 (1-3-3-2-5-7):

A3-1	Create claim form	(Insured)
A3-2	Enter claim form details into the system (incomplete)	(Employee)
A3-3	Start the process "Check for completeness"	(Employee)
A3-4	Contact the insured party to complete the details	(Employee)
A3-5	Enter claim form details including one additional detail into the system (incomplete)	(Employee)
A3-6	Start the "Check for completeness" process	(Employee)
A3-7	Contact the insured party to complete details	(Employee)
A3-8	Enter claim form details including additional details into the system (complete, claim sum > €2,500)	(Employee)
A3-9	Start "Check for completeness" process	(Employee)
A3-10	Start "Determine number of claims and claim sum" process	(Employee)
A3-11	Assess notification of claim	(Head)
A3-12	Claim is accepted	(Head)
A3-13	Start "Create letter" (approval letter) process	(Employee)
A3-14	End process	

The tester has described the physical formulation of TC3 (1-3-3-2-5-7) as follows:

Insured party:
A3-1 Create claim form with following details:

Name	: Janssen, J.
Address	: Amsterdamstraat 7, Utrecht
Date of loss	: <empty>
Description of loss	: <empty>
Cause of loss	: Theft from home
Claim sum	: €4.250

Employee (EMP_01):

A3-2	Enter claim form details into the system (without "Date of loss" and without "Description of loss")
A3-3	Start "Check for completeness" process (form is incomplete)
A3-4	Contact insured party to obtain "Date of loss"
A3-5	Enter claim form details in the system ("Date of loss" is 1 December 2006)
A3-6	Start "Check for completeness" process (form is incomplete)
A3-7	Contact insured party to obtain "Description of loss"
A3-8	Enter claim form details in the system ("Description of loss" is "Stolen necklace")
A3-9	Start "Check for completeness" process (form is complete)
A3-10	Start "Determine number of claims and claim sum" process (claim sum > 2,500, therefore assessment by HoD)

Head of Department (HEAD_01):

A3-11	Assess notification of claim
A3-12	Accept claim

Employee (EMP_01):

A3-13	Start "Create letter" process (approval letter is created)
A3-14	End process

Data-driven testing: Pairwise testing

Verify that every pair of input parameters is tested at least once



A

10

×

B

10

×

C

10

×

D

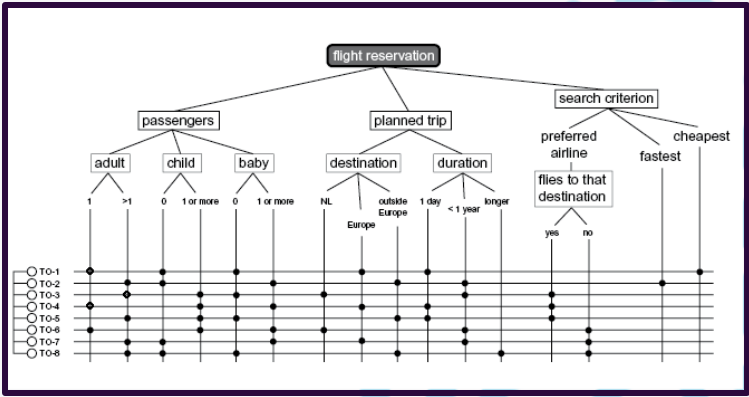
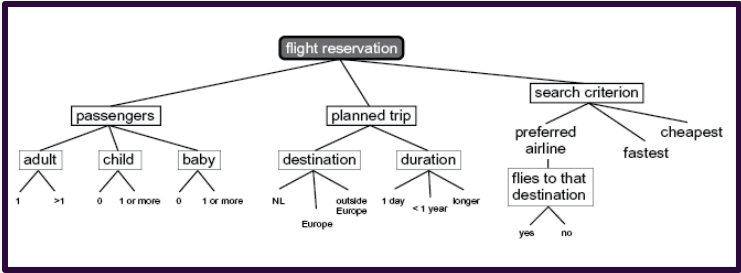
10

×

E

10

2⁸ Password Criteria
2³ Input Fields
2 Button State
2⁸ Error Message
2⁸ Visual Indicator
=
1,048,576 combinations



Example

In order to illustrate the principle, 4 test cases are made physical in the figure below. With every test case, the physical values are defined for all the necessary input data and the predicted result is described in concrete terms.

	TG-1	TG-2	TG-3	TG-6
Customer name	Jansen	Bruegel	Voort	Hansma
#Adults	1	2	3	1
#Children	0	0	1	4
#Babies	0	2	0	1
Destination	France-CDG	Singapore	The Netherlands Eindhoven Airport	The Netherlands Eindhoven Airport
Departure date	12-02-2006	14-02-2006	15-02-2006	16-02-2006
Return date	12-02-2006	15-02-2006	15-04-2006	23-02-2006
Search criterion	Cheapest	Fastest	KLM	Senegal Airlines
Predicted result				Message: "Airline does not fly to chosen destination"
Airline	Korean Air	Canada Air	KLM	
Flight number	KA0455	CA0833	KL1288	
Price	€ 44	€ 865	€ 83	

Condition-driven testing: Modified Condition / Decision Coverage

Verify that every possible outcome of a condition is the determinant of the outcome of the decision at least once.



IF password meets all criteria THEN
create account successfully ELSE
display appropriate error message

```
IF password_length >= 12
AND password_contains_uppercase = TRUE
AND password_contains_lowercase = TRUE
AND password_contains_digit = TRUE
AND password_contains_special_character = TRUE
AND password_contains_username = FALSE
AND password_contains_email = FALSE
AND password_is_common = FALSE

THEN password_validation_status := 'Valid'

ELSE
  IF password_length < 12 THEN
    display_error_message := 'Your password must be at least 12 characters'
  ENDIF

  IF password_contains_uppercase = FALSE THEN
    display_error_message := 'Password must contain at least one uppercase letter'
  ENDIF
ENDIF
```

Test Case ID	Password_Length >=12	Includes_Uppercase	Includes_Lowercase	Includes_Digit	Includes_Special_Ch
TC1	True	True	True	True	True
TC2	False	True	True	True	True
TC3	True	False	True	True	True
TC4	True	True	False	True	True
TC5	True	True	True	False	True
TC6	True	True	True	True	False
TC7	True	True	True	True	True
TC8	True	True	True	True	True
TC9	True	True	True	True	True

Test Case ID	Password_Length >=12	Includes_Uppercase	Includes_Lowercase	Includes_Digit	Includes_Special_Character	NOT Contains_Username	NOT Contains_Email_Address	NOT Is_Common_Password	Expected Result	Actual Results	Password
TC1	True	True	True	True	True	True	True	True	Accepted		P@ssw0rd1234!
TC2	False	True	True	True	True	True	True	True	Rejected		P@ssw0rd1!
TC3	True	False	True	True	True	True	True	True	Rejected		p@ssw0rd1234!
TC4	True	True	False	True	True	True	True	True	Rejected		P@SSWORD1234!
TC5	True	True	True	False	True	True	True	True	Rejected		P@ssword!!!!
TC6	True	True	True	True	False	True	True	True	Rejected		P4ssw0rd1234567
TC7	True	True	True	True	True	False	True	True	Rejected		P@ssw0rd1234!user
TC8	True	True	True	True	True	True	False	True	Rejected		P@ssw0rd1234!email@example.com
TC9	True	True	True	True	True	True	True	False	Rejected		Password1234!

Requirement creation

Supports the extraction and structuring of various types of requirements and ensures detailed, structured outputs that adhere to industry standards.

1

Input
(choose one)

Transcript

Code

Microsoft Word Document

Microsoft Word Document

2

Specifics

3

Knowledge

Microsoft Word Document

4

API call

5

Output

Generated requirements

Microsoft Word Document

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for Software & Quality Engineering

Requirement Creation

As a Project Team Member I want to be assisted with the creation of a detailed requirements document based on the conversation document.

Choose Model:
OpenAI GPT-4o Preview

Upload File Enter Text

Upload File here:

1b_REQ-transcript password.docx

Select Target Audience:
Business Analyst

Select Tone:
Strictly Formal

Choose Requirement Type:
Functional

Additions of User-specific behavior and any reference material (Optional):
No Additional Criteria: Do not introduce any new requirements, criteria, considerations, or expansions beyond those explicitly stated in the provided transcript.

Clear

Reg management:
Existing Custom

Upload reg file here:

1b_RAC_modern password security for system designers.pdf

Process Your RAC

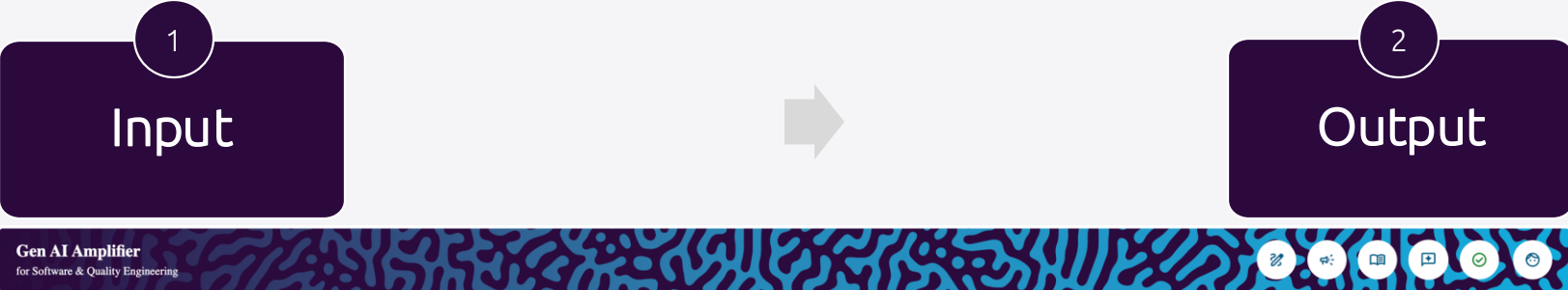
Reiterate on your output...

2. Functional Requirements Table

Requirement ID	Requirement Type	Priority	Business Process	Detailed Description	Acceptance Criteria	Function Points
FR-1	Functional	High	Account Creation, Password Update, Password Reset	The system shall enforce a minimum password length of 12 characters for all user passwords.	Users are prevented from setting passwords shorter than 12 characters during account creation, password updates, and password resets. Testing confirms that passwords less than 12 characters are rejected with a clear and specific error message.	3 (Simple validation logic)
FR-2	Functional	High	Account Creation, Password Update, Password Reset	The system shall require at least one uppercase letter in user passwords.	Users are prevented from setting passwords that do not contain at least one uppercase letter. Testing confirms that such passwords are rejected with a clear and specific error message.	3 (Simple validation logic)
FR-3	Functional	High	Account Creation, Password Update, Password Reset	The system shall require at least one lowercase letter in user passwords.	Users are prevented from setting passwords that do not contain at least one lowercase letter. Testing confirms that such passwords are rejected with a clear and specific error message.	3 (Simple validation logic)
FR-4	Functional	High	Account Creation, Password Update, Password Reset	The system shall require at least one digit in user passwords.	Users are prevented from setting passwords that do not contain at least one digit. Testing confirms that such passwords are rejected with a clear and specific error message.	3 (Simple validation logic)
FR-5	Functional	High	Account Creation, Password Update, Password Reset	The system shall require at least one special character from the set !@#\$%^&*~ in user passwords.	Users are prevented from setting passwords that do not contain at least one specified special character. Testing confirms that such passwords are rejected with a clear and specific error message that lists the allowed special characters.	3 (Simple validation logic)
FR-6	Functional	High	Account Creation, Password Update, Password Reset	The system shall prevent users from setting passwords that contain any substring of three or more consecutive characters from their username, case-insensitive.	Users are prevented from setting passwords containing any substring of three or more consecutive characters from their username, regardless of case. Testing confirms that such passwords are rejected with a clear and specific error message.	4 (Moderate validation complexity)
FR-7	Functional	High	Account Creation, Password Update, Password Reset	The system shall prevent users from setting passwords that contain any substring of three or more consecutive characters from their email address (both local part and domain), case-insensitive.	Users are prevented from setting passwords containing any substring of three or more consecutive characters from their email address, regardless of case. Testing confirms that such passwords are rejected with a clear and specific error message.	4 (Moderate validation complexity)
FR-8	Functional	High	Account Creation, Password Update, Password Reset	The system shall prevent users from setting passwords that are included in a provided list of common passwords.	Users are prevented from setting passwords that match any password in the common passwords list. Testing confirms that such passwords are rejected with a clear and specific error message.	4 (Moderate complexity due to list lookup)
FR-9	Functional	High	Account Creation, Password Update, Password Reset	The system shall display all password validation errors simultaneously when multiple criteria are not met.	Users receive all relevant error messages at once, allowing them to address all issues simultaneously. Testing confirms that multiple validation errors are displayed together when applicable.	2 (Simple UI logic)
FR-10	Functional	High	User Interface	The system shall provide clear and helpful error messages in plain language without technical jargon.	Error messages clearly explain the validation issues and guide users on how to create a compliant password. Testing	2 (Content design)

Refinement analysis

Detects ambiguities, refines input, and ensures consistency in requirement documents through readability scoring, term alignment, and tone analysis.



Generated requirements



Approximately 2,500 words

Gen AI Amplifier for Software & Quality Engineering

Requirement Refinement

As a Product owner, to enhance user stories, I want to be assisted refinement of requirements document created.

Choose Model:
OpenAI GPT-o1 Preview

Upload File | Enter Text

Upload file here:
1e. REQ - output.docx

Select Target Audience: Software Engineer | Select Tone: Strictly Formal

Select Language Fluency: Advanced

Addition of User-specific behavior and any reference material (Optional):

Reset | Generate

Output

1. Executive Summary

Total Ambiguities Found: 17

Ambiguity Summary:

Ambiguity Type	High Impact	Medium Impact	Low Impact	Total
Lexical	0	0	0	0
Referential	0	0	0	0
Syntactic	1	0	0	1
Vague	2	11	3	16
Total	3	11	3	17

Input Confidence Factor: 7/10
(Moderately clear but with significant ambiguities)

Readability Score: Grade Level 13 (Suitable for ages 18-19)

Input Confidence Factor: 7/10

Total Words Analyzed: Approximately 2,500 words

2. Detailed Ambiguity Table

Ambiguity ID	Ambiguous Term/Phrase	Occurrences & Context	Ambiguity Type	Impact Level	Suggested Rephrasing
A1	"comprehensive password policy"	Summary, Paragraph 1	Vague	Medium	Specify the exact components of the password policy: "The system requires the implementation of a password policy that enforces specific security criteria."
A2	"clear and helpful error messages"	Summary, Paragraph 1; FR-09	Vague	Medium	Define guidelines for error messages: "The system must provide error messages that are easily understandable and offer specific instructions for correcting the error."
A3	"ensure efficient performance without compromising security"	Summary, Paragraph 1	Vague	Medium	Specify performance benchmarks: "The system must validate passwords within 500 milliseconds to ensure responsiveness without compromising security."
A4	"provide a solution of these issues"	FR-06	Syntactic	High	Clarify the action and scope: "The system shall reject requests that contain errors."

Reiterate on your output...

3

Refined Output

Refined requirements



User stories

Convert complex project requirements into detailed, actionable user stories with corresponding acceptance criteria



Refined requirements



Gen AI Amplifier
for Software & Quality Engineering

Requirements to User Story G...

As a Product owner, to enhance user stories, I want to be assisted while defining and reviewing my user stories

Choose Model:
OpenAI GPT-o1 Preview

Upload Requirement File Enter Text

Upload File here:
3. REF-password-refined output.docx

Select user story type:
Azure DevOps Jira

Addition of User-specific behavior and any reference material: (Optional):
No Additional Criteria: Do not introduce any new requirements, criteria, considerations, or expansions beyond those explicitly stated in the provided input. Make every effort to consolidate the user stories

Clear

Reset Generate

Output

3686 - Plan

Requirements to User Story Generation
Version: v0

UserStory_ID	Requirement_ID	Epic	Feature	User Story Title	User Story Description	Acceptance Criteria
US001	FR-01	User Account Security Enhancements	Password Policy Enforcement	Enforce minimum password length of 12 characters	As a user, I want to create a password with at least 12 characters so that my account is more secure.	- Passwords shorter than 12 characters are rejected.-br> Given the user attempts to set a password shorter than 12 characters,-br> when they submit the form,-br> then the system displays an error message: "Your password must be at least 12 characters long."-br> The system accepts passwords that are 12 characters or longer.-br> Password validation completes within 500 milliseconds.-br> Validation is consistent across account creation, password updates, and resets.
US002	FR-02	User Account Security Enhancements	Password Policy Enforcement	Require at least one uppercase letter in passwords	As a user, I want to include uppercase letters in my password so that it meets security requirements.	- Passwords lacking an uppercase letter are rejected.-br> Given the user enters a password without an uppercase letter,-br> when they submit the form,-br> then the system displays an error message: "Your password must include at least one uppercase letter."-br> The system accepts passwords containing at least one uppercase letter.-br> Password validation completes within 500 milliseconds.-br> Validation is consistent across all password functionalities.
US003	FR-03	User Account Security Enhancements	Password Policy Enforcement	Require at least one lowercase letter in passwords	As a user, I want to include lowercase letters in my password so that it meets security requirements.	- Passwords lacking a lowercase letter are rejected.-br> Given the user enters a password without a lowercase letter,-br> when they submit the form,-br> then the system displays an error message: "Your password must include at least one lowercase letter."-br> The system accepts passwords containing at least one lowercase letter.-br> Password validation completes within 500 milliseconds.-br> Validation is consistent across all password functionalities.
US004	FR-04	User Account Security Enhancements	Password Policy Enforcement	Require at least one digit in passwords	As a user, I want to include digits in my password so that it meets security requirements.	- Passwords lacking a digit are rejected.-br> Given the user enters a password without a digit,-br> when they submit the form,-br> then the system displays an error message: "Your password must include at least one digit."-br> The system accepts passwords containing at least one digit.-br> Password validation completes within 500 milliseconds.-br> Validation is consistent across all password functionalities.
US005	FR-05	User Account Security Enhancements	Password Policy Enforcement	Require at least one special character in passwords	As a user, I want to include special characters in my password so that it meets security requirements.	- Passwords lacking a special character from the set !@#\$%^&*()_ are rejected.-br> Given the user enters a password without a required special character,-br> when they submit the form,-br> then the system displays an error message: "Your password must include at least one special character (!@#\$%^&*()_)." -br> The system accepts passwords containing at least one specified special character.-br> Password validation completes within 500 milliseconds.-br> Validation is consistent across all password functionalities.
US006	FR-06	User Account	Password Policy	Prevent use of user-	As a user, I want the system to	- Passwords containing any sequence of three or more consecutive characters from the user's

Push to Azure DevOps Save as Copy

Reiterate on your output...

User stories

Sogeti Gen AI Amplifier for Software & Quality Engineering

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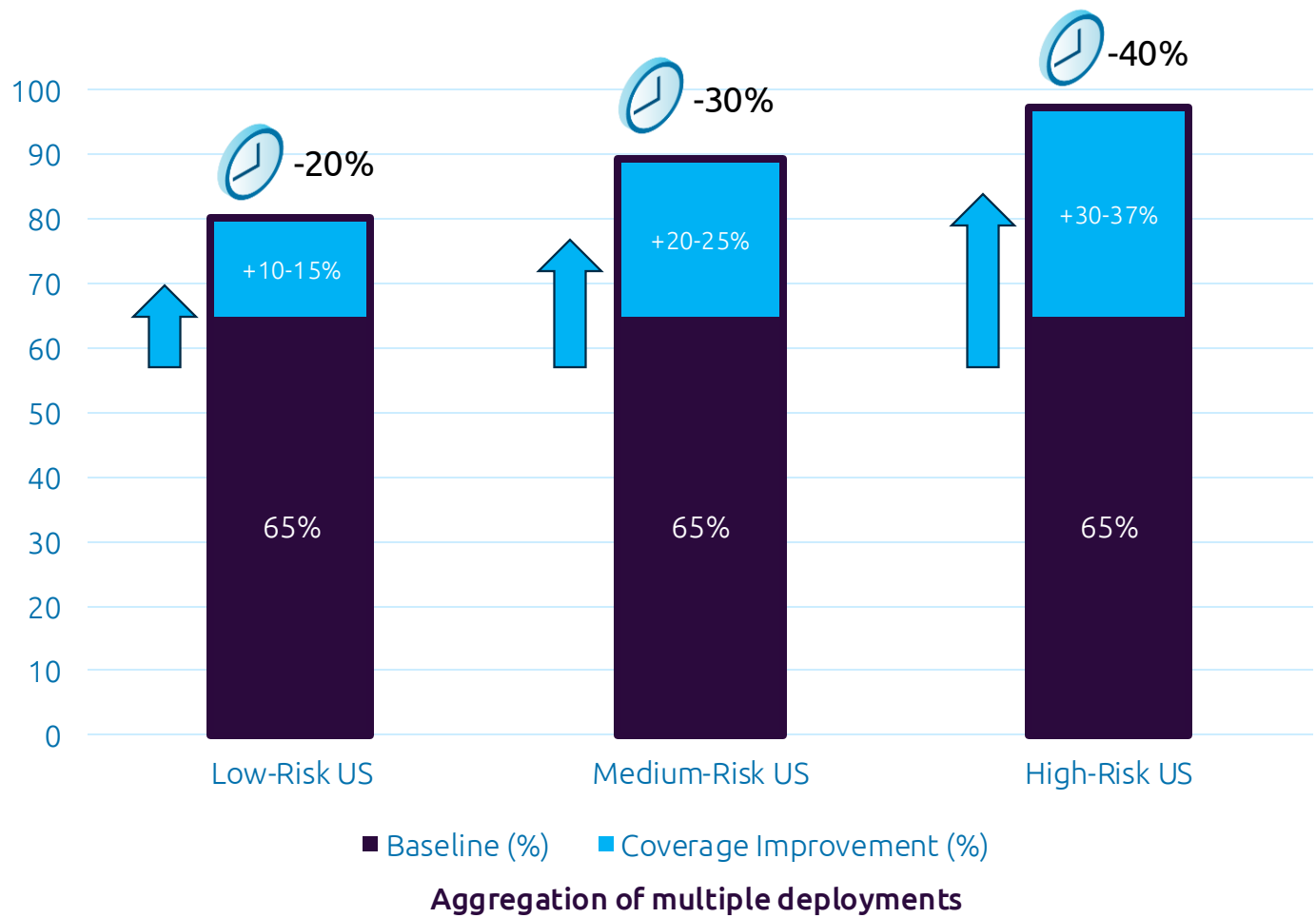
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Section 3

Lessons learned

Productivity gain on Test Design



The higher the risk, the greater the improvement in effectiveness and efficiency

Observations

- 1** The mandate for 'Expert in the loop'
- 2** Focus on the end-to-end process (e.g. high-quality input)
- 3** Productivity gains depend heavily on the context
- 4** The need to facilitate custom use cases
- 5** Integrate within the broader ecosystem

Observations

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AI enhances human intelligence but cannot replace domain expertise

7

Output quality depends on large language models

8

Upfront time is necessary to assess the quality of input assets

9

Requires more specialized expertise than before

10

Real productivity gains require real-world experimentation

